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RESEARCH ARTICLE

A Hybrid CNN-SVM Algorithm for Detecting Manufacturing Defects

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ABSTRACT In complex industrial manufacturing processes, various defects may occur in products because of the poor quality of the equipment, materials, and human factors. These defects may lead to undesirable situations due to reasons such as increasing costs, shortening service life, reducing customer satisfaction, and adversely affecting quality. To reduce these problems, it is crucial to detect products with such defects quickly and effectively. Therefore, within the scope of this study, a hybrid Convolutional Neural Network (CNN)-Support Vector Machine (SVM) hybrid model is introduced to improve the classification accuracy of production defects. The proposed framework incorporates a CNN for feature extraction, coupled with a Least Absolute Shrinkage and Selection Operator (LASSO) to facilitate feature selection. This approach aims to mitigate the dimensionality of the feature space and is complemented by the implementation of a SVM classifier. The performance of the model was tested on three widely used publicly available datasets: metal casting, textile fabric, and food images. Significant and high-performance results in both classification accuracy and execution time, which are important performance metrics for real-time applications, were obtained by the proposed hybrid CNN-SVM algorithm. Statistical analyses are also performed to test whether there is a significant difference between the CNN-SVM hybrid and conventional CNN and between CNN-SVM hybrid and the previous studies in literature. This study demonstrates that the AlexNet-SVM hybrid with LASSO outperforms the conventional CNN in terms of execution time. Furthermore, the AlexNet-SVM hybrid without LASSO achieved classification accuracies of 98,29%, 99,82%, and 97,37% for the casting, food, and fabric datasets, respectively.

INDEX TERMS Classification, convolutional neural network, deep learning, defect detection, hybridization, manufacturing defects, support vector machine.

I. INTRODUCTION

Manufacturing defects can occur during production in complex industrial processes. These defects can increase operating costs, shorten the service life of products, negatively affect quality, cause manufacturing waste, and lead to customer dissatisfaction. These defects may be caused by manufacturing equipment, material failures, human factors, poor working conditions, or the improper use of technology. Detecting and reducing defects in manufacturing processes are crucial for efficiency, quality, and cost control. Customers always demand high-quality products and companies that

succeed in this regard can gain greater customer loyalty. In this context, it is important to detect defects more quickly and accurately and to take measures to prevent or reduce them. Rapid detection of defects can improve product quality and prevent potential problems. Early detection can reduce recall and remanufacturing costs, which in turn reduces operating costs. High-quality products with low defect rates can increase a company's competitiveness and market share. Therefore, companies must develop core competencies in defect detection to improve the quality of the products without hindering manufacturing [1].

Among the product defects, poor product quality is most often observed in surface defects. Therefore, defect detection on product surfaces is significant criterion [2]. Defects

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are generally defined as deficiencies or variations in areas that differ from those of a normal sample. In this study, Deep Learning (DL) methods were used to detect and reduce product surface defects to improve industrial manufacturing processes.

DL methods offer the ability to analyse large datasets and predict future defect probabilities. In the era of Industry 4.0, the automation of manufacturing systems and data analytics are of great importance [3]. DL methods are popular for improving defect detection and reduction processes in a manner that is suitable for this era [4]. Artificial intelligence and DL can be successful in large and complex datasets owing to their ability to work with such data. They also have the capability to recognize complex and hidden relationships that are crucial for discovering defects in manufacturing processes because these defects may be based on hidden patterns that are difficult to detect using conventional methods. Thus, they are beneficial for the detection and analysis of defects in the manufacturing processes [5]. DL methods have a customizable structure based on the requirements of a specific manufacturing system and offer an adaptable solution to meet the needs of a particular company or industrial process. These methods can learn from data, making it easier for companies to adapt to constantly changing industrial processes. They also offer automatic feature extraction and allow companies to extract more information from data [6].

Methods based on image processing derive inferences from image detection, image processing, defect detection, and classification. Image processing techniques generally define objects based on their colour, texture, or geometric features. This is one of the key technologies for automating defect detection and classification processes without the need for time consuming human intervention. It offers better accuracy and lower costs than conventional methods and can significantly improve efficiency.

DL methods are a type of Machine Learning (ML) technique derived from Artificial Neural Networks (ANNs) that can automatically extract feature representations from raw data. In most cases, unlike conventional ML methods, data preprocessing is not necessary. Instead, DL methods learn characteristic hierarchies with higher hierarchical features by combining the lower level features [7]. Thus, they have been successfully used to solve complex, high dimensional problems. Convolutional Neural Networks (CNN) are among the most successful DL methods [8].

DL is a rapidly developing technology that has achieved significant success in object detection [9], intelligent robotics [10], sound detection and classification [11], smart city security applications [12], [13], acoustic recognition [14], and aerospace [15], [16]. DL has a deep neural network (DNN) structure with multiple convolutional layers. By combining low level features, a more abstract, high-level representation of features or categories can be formed, enabling the data to be better understood in terms of abstract elements such as edges and shapes. This can improve the

efficiency of deep learning algorithms [17], [18]. On the other hand, CNNs are used in a wide range of computer vision tasks, such as face recognition, pedestrian identification, text detection in images, and target tracking. Several studies have explored the latest techniques, applications, and other aspects of deep learning methods for defect detection, including ultrasonic inspection and machine vision [19], [20], [21]. Surface defect detection methods are increasingly being studied, and the number of studies in the literature continues to increase [22]. Different types of surface defect detection studies have been conducted for various applications such as fabric defects [23], metal defects [24], [25], and other industrial applications [26].

Ren et al. [27] proposed a deep learning algorithm to detect various types of defects using an imbalanced training sample pool for Printed Circuit Board Assembly (PCBA) manufacturing. In this method, an overall defect recognition accuracy of 98% was achieved on PCBA images using a novel batch sampling method and a sample-weighted cost function.

Masci et al. [28] designed a CNN architecture with five layers to classify surface defects. Experimental studies were conducted using a dataset containing seven different defect types with the aim of optimizing the proposed CNN architecture. According to these results, the proposed CNN architecture outperformed only SVM based approaches supported by different features.

Weimer et al. [29] developed several CNN architectures with between 5 to 11 layers for surface defect detection. These architectures were analysed using synthetic surface images that contained six different patterns and defect types. The results show that the CNN-based methods overpassed traditional methods.

Jing et al. [30] proposed a CNN-based model for detecting fabric defects. In the proposed model, fabric images were divided into sections, and each section was labelled. In the experimental studies, an average accuracy score of 97.31% was obtained.

Yi et al. [31] developed a 14-layer CNN architecture for classifying steel surface defects. In the proposed architecture, the fully connected layers follow the convolutional and pooling layers. The network architecture was designed to categorize six different image defects. In experimental studies, the Northeastern University (NEU) surface dataset was used, and the model achieved an accuracy score of 99.05%.

Cheon et al. [32] developed a system that combines K-nearest neighbours (KNN) and CNN methods. In this system, wafer surface defects were classified using images from a scanning electron microscope. In the experiments, the authors' dataset was used, and a classification accuracy of 96.2% was achieved.

Masci et al. [33] proposed a multiscale pyramidal pooling network architecture for classifying surface defects in steel. The proposed network architecture includes a pyramidal pooling layer that is capable of processing images of various sizes. In the pyramidal pooling layer, images of different sizes

are resized to a fixed size using pooling and convolution layers. The fixed-size images were classified using the CNN architecture. In experimental studies, the proposed model produced effective results and achieved higher scores than the models used in previous studies.

Yanan et al. [34] used the YOLOv3 model for rail surface defect detection. In this study, rail surface images were divided into two parts. The YOLOv3 model is applied separately to each part. A dataset consisting of rail images created by the authors was used in the experimental studies. Consequently, a defect detection success rate of over 97% was achieved.

Yuan et al. [35] developed a network architecture called MOLO, which combines the MobileNet V2 and YOLOv3 network architectures. In the proposed model, the feature maps obtained from the MobileNet V2 network architecture are combined with the YOLOv3 network structure. In the experimental study, it was determined that the proposed MOLO architecture achieved higher scores than YOLO architecture.

Li et al. [36] proposed an architecture that combines the Single Shot MultiBox Detector (SSD) and MobileNet architectures for real-time defect detection. In the proposed approach, the SSD architecture is optimized for defect detection. The developed model was applied to detect surface defects, such as gaps, cavities, burrs, and abrasions on glass bottles in the production line. In experimental studies, the proposed model was compared with CNN architectures and conventional surface defect detection models, and it was found that the proposed model outperformed others.

Jeong et al. [37] introduced the “Hybrid-DC” model, a hybrid of ResNet-50 and Vision Transformer, for classifying surface defects in the steel rolling process. They reported an accuracy of 99.44%, demonstrating statistically significant improvements over conventional CNNs due to the hybrid attention layer and attention fusion.

Hussain et al. [38] proposed a CNN-ML hybrid where deep features extracted from VGG19 were mapped to an SVM classifier for steel strip defect detection. Their model achieved 99.79% accuracy on the NEU-CLS dataset and 99.64% on XSDD, along with high precision, sensitivity, and F1 scores.

Wang et al. (2024) [39] addressed the issue of information density imbalance in integrated circuit surface defect detection using a deep learning-based approach, reporting effective results in industrial applications.

Parlak and Emel (2025) [40] developed a two-stage method to detect both the type and amount of internal defects in high-pressure aluminium castings in real time, achieving a mAP value of 0.971 with the YOLOv5 algorithm.

A comprehensive review published in 2025 [41] provided detailed coverage of fault detection, AI-based methodologies, and future research directions in the context of Industry 4.0.

In this study, convolutional neural networks, which are widely used DL methods in the literature, were used as feature extractors and Support Vector Machine hybridization was performed as the classifier layer. The advantages and disadvantages of using the LASSO method as a feature selector in the CNN-SVM hybrid were also examined and its contribution to the hybrid was evaluated. The majority of hybrid combinations designed and evaluated within the scope of this study stand out in terms of time and accuracy compared to conventional methods. In this study, a CNN-SVM hybrid model was developed to increase the performance of traditional CNN network structures. The features obtained from the images given as input to the CNN were used as input for the SVM classifier. In addition, a significant performance increase was achieved in terms of execution time when the LASSO feature selection method was used in the proposed hybrid algorithm. Using the proposed methods, defects were classified in three datasets with different product types (casting, food and fabric). The proposed Hybrid CNN-SVM method was implemented based on four different CNN architectures (AlexNet, MobileNet V2, ResNet 50 and ShuffleNet) and the obtained results showed that it performed better than the classification operations where conventional CNN architecture was used. As a result of the experimental studies, it was shown that using CNN as a feature extractor and SVM as a classifier contributed to the classification success and it was revealed that reducing the number of the features extracted from CNN with the LASSO feature selection algorithm made a significant contribution to the execution time. In addition, the Kolmogorov Smirnov and Shapiro Wilk tests were used to examine whether the 30 execution results made on three data sets were distributed normally. The one sample t test was used to examine and interpret whether the differences were significant compared to the results obtained from other methods in the literature and also from the conventional CNN.

In the field of defect detection in manufacturing, computer vision-based methods have become increasingly prevalent, with deep learning models such as CNNs demonstrating high accuracy levels. However, CNNs alone face persistent challenges, including high computational cost, susceptibility to overfitting, limited interpretability, and weak generalizability across domains, which restrict their deployment in real-time industrial environments. Moreover, the softmax function, typically used in CNN classifiers, often fails to construct sufficiently sharp decision boundaries in imbalanced datasets, leading to misclassification of minority defect types despite high overall accuracy [42].

Support Vector Machines (SVMs), on the other hand, are capable of generating sharper decision boundaries through margin maximization, enabling more reliable separation of smaller sample groups and thereby producing more balanced outcomes compared to softmax-based classifiers. Yet, when applied directly to raw image data, SVMs encounter

severe computational burdens due to the curse of dimensionality [43]. In other words, while CNNs are limited in flexibility at the classification stage, SVMs alone struggle with efficiency in high-dimensional feature spaces.

This study addresses these limitations by proposing a hybrid CNN-LASSO-SVM framework designed to exploit the complementary strengths of each method. Specifically:

1. CNNs serve exclusively as an automatic feature extractor, providing high-dimensional and semantically rich feature representations.
2. LASSO regularization performs dimensionality reduction by eliminating redundant or less informative features, thereby improving both efficiency and generalization.
3. SVMs then act as the final classifiers, constructing sharper decision boundaries and enhancing robustness to class imbalance.

Compared to existing hybrid approaches the proposed framework makes three distinct contributions: (i) a modular three-stage integration that enhances interpretability and scalability, (ii) a diversified validation strategy across four CNN architectures (AlexNet, MobileNet V2, ResNet-50, ShuffleNet) and three industrial datasets (casting, fabric, and food), which mitigates architecture- and domain-dependency, and (iii) a comprehensive performance analysis incorporating accuracy, F1-score, sensitivity, specificity, execution time, and statistical significance tests over 30 experimental runs.

In summary, the proposed method addresses a well-defined research gap: the lack of hybrid defect detection models that are simultaneously computationally efficient, generalizable across domains, and validated through diversified statistical evidence. The results demonstrate that the hybrid CNN-SVM with LASSO significantly improves both efficiency and classification balance, making it suitable for real-time industrial defect detection systems.

II. BRIEF OVERVIEW OF EMPLOYED ALGORITHMS

In this section, a brief overview of CNN and SVM architectures, as well as feature extraction, and selection methods, is provided.

A. CONVOLUTIONAL NEURAL NETWORKS

The structure of CNNs was inspired by neurons in human and animal brains. CNNs were primarily used for image recognition and computer vision tasks. CNNs consist of three main types of layers: convolutional filters, pooling layer, and fully connected layer. CNNs are similar to conventional neural networks in terms of they consist of neurons that self-optimize through learning [44]. Each neuron receives an input and performs an operation (e.g. scalar multiplication followed by a nonlinear function), which forms the basis of neural networks. The entire CNN is represented by weights that map raw input image vectors to the final output class scores. The last layer contains loss functions associated with the classes. A CNN architecture is formed when the lay-

ers are stacked. A simplified CNN architecture is shown in Figure-1 [45].

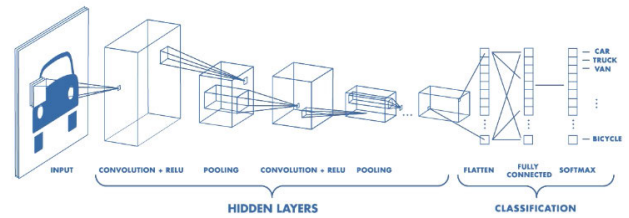


FIGURE 1. Simplified CNN structure.

The input layer contains the pixel values of the image. The convolution layer determines the output of the neurons connected to the local regions of the input by computing the scalar product of their weights and the corresponding region of the input volume. An activation function, such as a sigmoid or rectified linear unit (ReLU) function is applied to the input which is the output produced by the previous layer. The pooling layer then performs subsampling along the spatial dimension of the given input, further reducing the number of parameters in this activation.

Convolution is an integral operation that expresses how a function is influenced by another function. The convolution layer plays a vital role in the functioning of CNNs. Convolution also promotes parameter sharing in a deep neural network architecture. The convolution layers significantly reduce the complexity of the model by optimizing the output.

In this study, four image classification architectures were used within the study framework. Each architecture has its own advantages and disadvantages. The success rate of each architecture varies depending on the dataset. The CNN architectures used in this study are as follows:

- 1) AlexNet Architecture [46]: AlexNet is a convolutional neural network with eight layers. This CNN architecture was trained on more than one million images from the ImageNet database. AlexNet uses ReLU activation functions, which allow it to learn faster. The main layers of the AlexNet architecture consist of convolution, pooling, and fully connected layers. The AlexNet architecture was designed to classify 1000 objects, and the error rate in object recognition was reduced from 26.2% to 15.3%.
- 2) MobileNet V2 Architecture [47]: MobileNet V2 is a model trained on a subset of the ImageNet database. The model was trained on over a million images and was able to classify these images into 1000 object categories. It is widely used because it is lightweight compared to other transfer learning models and can be deployed on mobile devices. MobileNet V2 is an architecture with fewer weights compared to MobileNet V1. Additionally, it is possible to perform pointwise convolution using 1×1 filters. The model's performance improved with these innovations.

- 3) ResNet 50 Architecture [48]: ResNet consists of 50 layers and has a deep structure. Its structure is different from that of conventional sequential network architectures, such as AlexNet. ResNet differs from other architectures in its own microarchitecture module structure. The most distinctive feature of ResNet is its direct connections, known as residual connections. It learns the difference between the input and output by adding the input directly to the output of the next layer, rather than the output of a typical layer. This prevents vanishing gradients and deepens the network. ResNet uses identity mapping to increase learning capacity and prevent overfitting as the network becomes deeper. The main feature of residual blocks is the equalization of input and output sizes, which facilitates network learning.
- 4) ShuffleNet Architecture [49]: ShuffleNet is an efficient convolutional neural network architecture designed for mobile and embedded devices with limited computational power. It aims to provide high efficiency in terms of computation and speed while maintaining competitive accuracy in tasks such as image classification. The basic idea behind ShuffleNet is to use pointwise group convolutions and channel mixing to reduce computational complexity without a significant loss in accuracy. The ShuffleNet architecture uses group convolution, which divide the input channels into groups and independently perform convolutions within these groups. This technique significantly reduces the number of parameters and hence reduces the computational cost. Furthermore, ShuffleNet, a strategy that optimizes the model further by decomposing standard convolutions into depth wise and pointwise operations, significantly reduces computation and model size without significantly affecting accuracy.

B. SUPPORT VECTOR MACHINE

A support vector machine is a supervised machine learning algorithm used for classification and regression. This algorithm separates data classes by determining an optimal hyperplane in a multidimensional feature space. This hyperplane is determined by maximizing the margin between the classes. For complex datasets with multiple classes and nonlinear boundaries, nonlinear hyperplanes can be generated using this algorithm by adjusting the SVM hyperparameters [50]. For this study SVM parameters are adopted from the referenced study [51]. In this referenced study linear kernel function with a scale of 1 and box constraint as 1 were used. SVM structure is shown in Figure-2.

C. LASSO FEATURE SELECTION

The Least Absolute Shrinkage and Selection Operator (LASSO) feature-selection method is widely used in machine learning to select a subset of relevant features from a larger feature set. It is particularly effective in high-dimensional

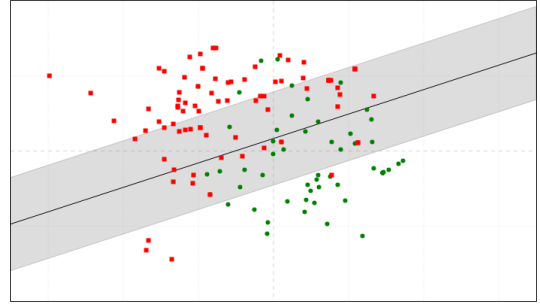


FIGURE 2. Simple SVM structure.

datasets where the number of features exceeds the number of examples. This method aims to reduce the complexity of the model by shrinking or minimizing the regression coefficients to zero, thereby creating a simpler and more consistent model based on a smaller number of important features.

LASSO is a linear model regularization technique that introduces a penalty term, known as L1 Regularization (or L1 Norm), into the least squares method. This penalty term increases the sum of the magnitudes of the coefficients, thereby shrinking some coefficients toward zero. Specifically, the penalty is defined as the sum of the absolute values of the feature coefficients, denoted by β . The strength of this penalty is controlled by a hyperparameter, λ (lambda). In this study, λ was empirically set to 0.05. The LASSO formulation is presented in Equation (1), where h_{real}^i represents the observed value for data point i , $h_{predicted}^i$ denotes the predicted value for data point i , and n is the total number of data points.

$$\frac{1}{n} \sum_{i=1}^n \left(h_{real}^i - h_{predicted}^i \right)^2 + \lambda \sum_{i=1}^n |\beta_i| \quad (1)$$

As the lambda value increases, the coefficients shrink further, and more features are reduced to zero. An appropriate lambda value must be selected to balance the complexity and accuracy of the model. There are several advantages and disadvantages of LASSO. Advantages include preventing overfitting by reducing the complexity of the model, automatically selecting features without the need for manual intervention to determine which features are important and increasing the interpretability of the model by creating simpler models based on fewer important features. Disadvantages include the possibility of selecting features that may not be important, the reduction in the likelihood of selecting correlated or noisy features, and difficulty in choosing the appropriate lambda value, which can significantly affect the performance of the model [52].

III. DEVELOPMENT OF HYBRID CNN-SVM ALGORITHM

In this study, a hybrid CNN-SVM algorithm is presented. To evaluate the performance of the CNN-SVM hybrid and the contribution of the hybridization process to the network structure, the conventional CNN was also trained and tested with the datasets used in this study. For the conventional

CNN, a simple partitioning process applied to the images in the datasets as 80% for training, 20% for testing. The classification results from the test data were used as performance metrics for the conventional CNN structure.

CNNs basically consists of three modules, backbone, neck and head. The backbone is responsible for extracting and encoding features from the input data. It acts as the core feature extractor, capturing low-level and high-level features from the input data. Neck module refines and aggregates features. The layer containing the training information (the last layer with learnable parameters) is called the network head. This layer is usually a fully connected layer or a convolutional layer, with an output size that matches the number of classes.

The network constructs a hierarchical representation of input images. Deeper layers contain higher-level features, constructed using the lower-level features of earlier layers [53].

To extract features for the CNN-SVM hybrid classification process, features were taken from the last layer of the backbone network specifically, the layer preceding the final learnable layer. This learnable layer corresponds to the last fully connected layer in all architectures. The specific layers from which features were extracted are as follows: the *dropout* layer for AlexNet and the *2D Global Average Pooling* layers for architectures such as MobileNetV2, ShuffleNet and ResNet50. Following feature extraction, LASSO feature selection was applied, and the selected features were classified using an SVM classifier.

For the CNN-SVM hybrid the images in the dataset were split into 80% training set and 20% testing set, and this process is repeated 30 times. For each repetition, 80% of the training data was fed into the backbone of the CNN. The 2-dimensional input data, passed through layers involving mathematical operations such as convolution, pooling and ReLU, and were transformed into a 1-dimensional vector at the flattening layer where the feature map is obtained. The SVM classifier was then trained and tested using these extracted and selected feature maps. To perform statistical analysis, the dataset was randomized, split, and processed 30 times. In this way, 30 different values for the performance metrics of the SVM classifier were obtained from 30 differently partitioned training and testing sets. The average and standard deviation of these 30 executions performance metrics constitute the final results.

Consequently, CNN backbone was used as a feature extraction method, and features were selected from the extracted features using the LASSO feature selection method. The SVM classifier was trained and tested using the selected features. The framework of the study is given in Figure-3.

Pseudo code for conventional CNN algorithm is given below.

BEGIN

1. Initialize CNN model (AlexNet, Resnet-50, MobileNet V2, ShuffleNet) with default parameters.
2. Load datasets (Casting, Fabric, Food).

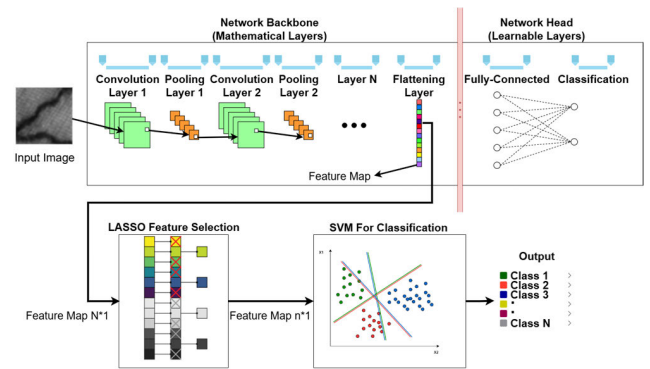


FIGURE 3. The framework of the study.

- Partition each dataset into training (80%) and testing (20%) sets using random partitioning
3. Train the initialized CNN model on the training set

4. Evaluate the trained model on the testing set.

5. Compute the performance metrics according to

Equations (2)-(6):

- Accuracy
- Sensitivity
- Specificity
- Precision
- F1 Score
- Execution Time

END

For the CNN-SVM hybrid algorithm, pseudo code is given below.

BEGIN

1. Initialize CNN models (AlexNet, Resnet-50, MobileNet V2, ShuffleNet) with default parameters.
2. Load datasets (Casting, Fabric, Food).

- Randomly partition each dataset into training (80%) and testing (20%) subsets.

3. Extract features:

- Train_Features = Feature_Extraction(CNN, Train_Data, Feature Layer)
Test_Features = Feature_Extraction(CNN, Test_Data, Feature Layer)

4. Perform LASSO feature selection:

Selected_Train_Features = LASSO(Train_Features)

Selected_Test_Features = Apply_Selected_Features(Test_Features, based on LASSO results)

5. Train and evaluate SVM classifier:

Train(SVM, Selected_Train_Features) Test(SVM, Selected_Test_Features)

6. Compute performance metrics using Equations (2)-(6):

- Accuracy
- Sensitivity
- Specificity

- Precision F1 Score
- Execution Time

6.Repeat Steps 2-6 until 30 independent executions are completed
END

IV. IMPLEMENTATION OF HYBRID CNN-SVM ALGORITHM

This section provides comprehensive information on the datasets used in the study, performance metrics, and statistical analyses.

A. DATASETS

Defect detection studies were conducted based on the different surface types found in the literature. All datasets consisted of image data, and each dataset has its own characteristics, as listed in Table-1.

TABLE 1. Dataset description.

	Casting	Food	Fabric
Number of Sample	1300	13599	25636
Number of Classes	2	6	5
Image Size	512 x 512	512 x 512	512 x 512
Colour Space	Gray	RGB	64 x 64 patch Gray

All datasets contain different types of data in terms of the number of samples, classes, image size, and colour space. The casting dataset consists of 2 classes, while the food and fabric datasets consist of 6 and 5 classes, respectively. All datasets contain images with a size of 512×512 , except for the fabric dataset, which is segmented into 64×64 patches. While the casting and fabric datasets are in grayscale, the food dataset is in the RGB colour space. The total number of data in the datasets varies between 1300 and 25636.

1) CASTING DATASET

When the studies in the literature are classified according to surface type, the surface type that has been studied the most is casting products. Casting is a manufacturing process in which a liquid material is poured into a mold, usually containing a hollow cavity of the desired shape, and is then allowed to solidify. Casting defects are undesirable in metal casting processes. The defects that may occur after the casting process are generally burrs, mold material defects, and air holes. The casting product dataset [54], contains images of cast manufacturing products and obtained as open access, consists of images related to the casting surface type. The dataset (Casting Product Image Data for Quality Inspection) contains 1300 images. These images have a size of 512×512 pixels in the grayscale colour space and are classified into two classes: defective and nondefective. Example images from the dataset, including a defective and a nondefective product sample, are shown in Figure-4.

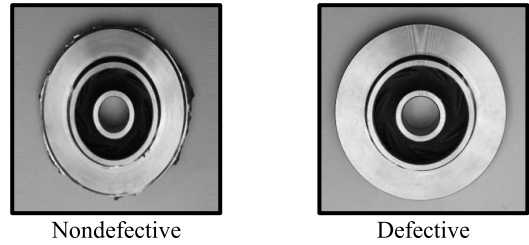


FIGURE 4. Casting dataset example images.

2) FOOD DATASET

The dataset named “Fresh and Rotten Fruits for Classification” [55] obtained as open access with a food data type, is an image dataset consisting of six classes: fresh and rotten classes of three different fruits. This dataset includes 2088 images in the fresh apple class, 1962 in the fresh banana class, 1854 in the fresh orange class, 2943 in the rotten apple class, 2754 in the rotten banana class, and 1998 in the rotten orange class. The dataset contains a total of 13599 images with 512×512 pixels in the RGB colour space. Example images containing a product sample from each class in the dataset are shown in Figure-5.

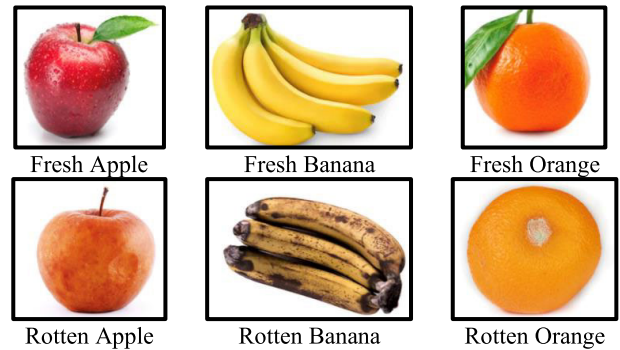


FIGURE 5. Food dataset example images.

3) FABRIC DATASET

The dataset with the fabric surface type, “Textile Texture Dataset for Fault Detection” (TILDA 400, 64×64 patches) [56] is designed for industrial fabric defect detection and classification. This dataset, obtained as open access, is based on the TILDA dataset. The images in the TILDA dataset are 768×512 pixels. These images were resized at 512×512 pixels and then separated into 64×64 pixel patches. The entire dataset consists of five classes and contains 25636 images. These classes are nondefective, thread defect, oil-stain, foreign object, and hole. Example images containing a product sample from each class in the dataset are shown in Figure-6.

B. PARAMETERS AND PERFORMANCE METRICS

The most important factor in the success of a network structure is the initialization of parameters. Correctly setting the

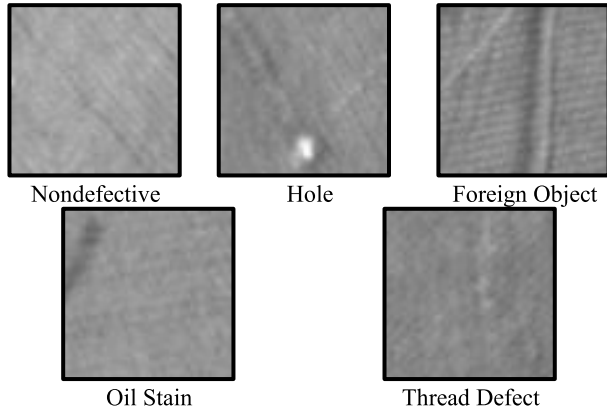


FIGURE 6. Fabric dataset example images.

initial parameters improves classification performance and helps the model converge to the desired outcome. To ensure a fair comparison of the CNN architectures used in this study, identical hyperparameters were applied across all datasets and architectures. This approach minimizes influence of varying hyperparameter settings and allows us to isolate and evaluate the impact of architectural differences on classification accuracy. The initial parameters used for CNNs are listed in Table-2.

TABLE 2. CNN parameters.

Parameters	Value
Epoch	6
Optimizer	SGDM
Batch Size	10
L2 Regularization	0.0001
Gradient Threshold	L2norm
Method	
Gradient Threshold	Inf
Shuffle	Every Epoch
Initial Learning Rate	0.0003

Small batch sizes allow the model to be updated more frequently and can enhance the generalization ability of the learning process. Therefore, a minimum batch size of 10 was selected as a balanced choice for both memory efficiency and generalization [57]. To avoid overfitting, the maximum number of training epochs was constrained. Previous studies have shown that CNN architectures often achieve high accuracies on small- and medium-sized datasets with a maximum of 5–10 training epochs [58]. In this study, the number of epochs was empirically determined as 6, since values greater than 6 did not improve accuracy and introduced unnecessary computational overhead. Learning rates that are too high may cause unstable convergence, while excessively low values can slow down training. The literature generally recommends values in the range of 0.0001–0.001 for small datasets [58]. Accordingly, we selected a learning rate of 0.0003, which

provided both stable convergence and an efficient training. Additionally, data shuffling was applied at each training step to prevent the model from learning data order, thereby enhancing generalization.

The success of the study was evaluated based on accuracy, precision, sensitivity, specificity, F1 score and execution time (in seconds) for all test data classifications. The true positive (TP), true negative (TN), false positive (FP), and false negative (FN) values were used to calculate these performance metrics [51].

The accuracy metric was calculated by dividing the number of correctly predicted positive and negative events by the total number of events. The formula for calculating the accuracy is as follows:

$$ACC = \frac{(TP + TN)}{TP + TN + FP + FN} \quad (2)$$

Sensitivity measures the ability of a model to correctly detect positive events. It is the percentage of accurately predicted positive events out of all the actual positive events. The formula used to calculate the sensitivity of the classification model is as follows:

$$SENS = \frac{TP}{TP + FN} \quad (3)$$

Specificity measures the ability of a model to correctly detect negative events. It is the percentage of accurately predicted negative events out of all the actual negative events. The formula used to calculate the specificity of the classification model is:

$$SPEC = \frac{TN}{TN + FP} \quad (4)$$

The precision metric determines the quality of positive predictions by measuring their correctness. It is the number of true positive outcomes divided by the sum of true positive and false positive predictions. The formula used to calculate precision is as follows:

$$PRE = \frac{TP}{TP + FP} \quad (5)$$

The F1 score is considered a better indicator of classifier performance than the standard accuracy measure. The F1 score is mathematically defined as follows:

$$F1 = 2 \times \frac{(Precision * Sensitivity)}{(Precision + Sensitivity)} \quad (6)$$

C. OBTAINED RESULTS

Using three different datasets, this section presents the results of hybridizing four different CNN architectures with an SVM classifier layer, both with and without LASSO feature selection. In addition to the hybrid architectures, the results obtained from conventional CNN architectures are also presented.

As a starting point, the number of features used by the SVM classifier -both with and without LASSO- is provided in Table-3. The use of LASSO aims to reduce the classification time for each data sample and decrease model complexity.

TABLE 3. Number of features with and without LASSO selection.

		Casting Dataset	Food Dataset	Fabric Dataset
AlexNet	CNN	4096	4096	4096
	Feature Size			
	LASSO	387	630	509
ResNet 50	CNN	2048	2048	2048
	Feature Size			
	LASSO	336	237	967
MobileNet V2	CNN	1280	1280	1280
	Feature Size			
	LASSO	366	234	268
ShuffleNet	CNN	544	544	544
	Feature Size			
	LASSO	235	126	262

The classification results with mean values and standard deviations for the casting dataset are shown in Tables 4-7. Results with better performance metrics are marked in bold in the tables.

When the results are examined, the classification performance in terms of accuracy is significantly improved for the hybrid AlexNet and ResNet 50 architectures without LASSO, and for the hybrid MobileNet V2 and ShuffleNet architectures with LASSO, compared to the results obtained using conventional CNN architectures.

When the classification accuracy of the conventional CNN architectures was examined, the AlexNet achieved the highest performance with an accuracy of 85.38%.

It is worth noting that, for the MobileNet V2 architecture, the sensitivity and specificity values are identical across multiple configurations. This is due to the nearly balanced dataset and the model's prediction behaviour, which yielded the same number of true positive and true negative cases.

For the Casting dataset, the CNN-SVM hybrid model that contributed most to classification accuracy was the hybrid version of AlexNet without LASSO, which increased the accuracy from 85.38% to 98.29%.

Using the proposed CNN-SVM hybrid instead of using conventional CNN significantly increases the classification success. Moreover, in all hybrid configurations both with and without LASSO- the execution time is considerably shorter than that of conventional CNN-based classification.

The classification results with mean values and standard deviations for the food dataset are shown in Tables 8-11. Results with better performance metrics are marked in bold in the tables.

For the Food dataset, all architecture combinations –both with and without LASSO- achieved higher classification accuracy compared to the conventional CNN.

Among the conventional CNN models, the highest classification accuracy was obtained using the AlexNet architecture, reaching 93.79%.

TABLE 4. Casting dataset results for AlexNet.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.8538	0.9829±0.0091	0.9692±0.0109
F1	0.8681	0.9858±0.0074	0.9741±0.0094
PRE	0.8013	0.9769±0.0129	0.9626±0.0159
SENS	0.9470	0.995±0.0077	0.9861±0.0119
SPE	0.7578	0.9651±0.0208	0.9453±0.0227
Time	6.792	0.0296±0.0379	0.0088±0.002

TABLE 5. Casting dataset results for ResNet 50.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.7462	0.9406±0.0136	0.9253±0.0134
F1	0.8235	0.9503±0.0116	0.9377±0.0116
PRE	0.9872	0.9430±0.0157	0.9346±0.0175
SENS	0.7064	0.9578±0.0152	0.9411±0.0179
SPE	0.9524	0.9155±0.0237	0.9022±0.0238
Time	17.3642	0.0245±0.0385	0.007±0.0011

TABLE 6. Casting dataset results for MobileNet V2.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.6538	0.6535±0.0214	0.6801±0.0239
F1	0.6382	0.6395±0.0234	0.6638±0.0249
PRE	0.6378	0.6403±0.0239	0.6642±0.0252
SENS	0.6387	0.6404±0.0226	0.665±0.0243
SPE	0.6387	0.6404±0.0226	0.665±0.0243
Time	15.6407	0.0232±0.0385	0.008±0.0017

TABLE 7. Casting dataset results for ShuffleNet.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.7077	0.7758±0.0228	0.7872±0.0189
F1	0.7889	0.8132±0.0221	0.8236±0.0185
PRE	0.9103	0.8131±0.0381	0.8268±0.0278
SENS	0.6961	0.8146±0.025	0.8213±0.0233
SPE	0.7500	0.7193±0.0447	0.7350±0.0354
Time	10.2833	0.0227±0.0356	0.007±0.0014

The best-performing method was the hybrid version of AlexNet with SVM, without LASSO, which increased the accuracy from 93.79% to 99.82%.

The SVM-based hybrid models using other CNN architectures also showed significant improvements in classification accuracy compared to their conventional counterparts. Furthermore, other performance metrics including F1 score,

precision (PRE), sensitivity (SENS), and specificity (SPE) also improved alongside accuracy (ACC). In addition to enhanced classification performance, reductions in execution time were observed. Overall, the CNN-SVM hybrid models developed for this dataset provide advantages across all evaluated performance metrics.

TABLE 8. Food dataset results for AlexNet.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.9379	0.9982±0.0003	0.9975±0.0004
F1	0.9374	0.9943±0.0011	0.9923±0.0013
PRE	0.9372	0.9942±0.0011	0.9922±0.0013
SENS	0.9395	0.9944±0.0011	0.9924±0.0012
SPE	0.9876	0.9989±0.0002	0.9985±0.0002
Time	3.1106	0.1324±0.0321	0.0476±0.0138

TABLE 9. Food dataset results for ResNet 50.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.9302	0.997±0.0004	0.9962±0.0006
F1	0.9287	0.9909±0.0011	0.9886±0.0017
PRE	0.9299	0.9907±0.0011	0.9885±0.0018
SENS	0.9322	0.991±0.0013	0.9887±0.0017
SPE	0.9861	0.9982±0.0002	0.9977±0.0003
Time	5.9543	0.0609±0.0096	0.0175±0.0007

TABLE 10. Food dataset results for MobileNet V2.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.9092	0.9829±0.0011	0.9824±0.0012
F1	0.9114	0.9484±0.0033	0.9472±0.0037
PRE	0.9109	0.9482±0.0033	0.9473±0.0038
SENS	0.9127	0.9487±0.0033	0.9473±0.0037
SPE	0.9816	0.9896±0.0007	0.9893±0.0008
Time	3.8890	0.0393±0.0049	0.0115±0.0005

The classification results with mean values and standard deviations for the fabric dataset are shown in Tables 12–15. Results with better performance metrics are marked in bold in the tables.

Upon examining the tables, it is observed that hybridizing the AlexNet and ShuffleNet architectures with SVM yields higher accuracy values without LASSO compared to their conventional CNN counterparts. In contrast, for the ResNet-50 and MobileNetV2 architectures, the CNN-SVM hybrid using LASSO achieves better classification performance than the conventional CNN. Among the conventional CNN models, the highest classification accuracy was achieved by

TABLE 11. Food dataset results for ShuffleNet.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.9232	0.9861±0.0009	0.9846±0.0012
F1	0.9216	0.9579±0.0027	0.9535±0.0036
PRE	0.9199	0.9576±0.0027	0.9529±0.0037
SENS	0.9254	0.9583±0.0029	0.9542±0.0035
SPE	0.9847	0.9916±0.0005	0.9907±0.0007
Time	2.2957	0.021±0.0049	0.0116±0.0008

TABLE 12. Fabric dataset results for AlexNet.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.9129	0.9737±0.001	0.9721±0.0009
F1	0.3186	0.5644±0.0151	0.5204±0.0156
PRE	0.3218	0.5056±0.0162	0.4519±0.0144
SENS	0.5476	0.6695±0.0161	0.6756±0.0204
SPE	0.9672	0.9687±0.0029	0.9754±0.0023
Time	2.0315	0.0738±0.0055	0.0141±0.0004

TABLE 13. Fabric dataset results for ResNet 50.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.9215	0.9629±0.001	0.972±0.001
F1	0.3818	0.5248±0.0232	0.5234±0.0147
PRE	0.3684	0.5115±0.0232	0.4649±0.013
SENS	0.4353	0.5438±0.0223	0.6454±0.02
SPE	0.9775	0.9137±0.004	0.9738±0.0023
Time	3.9234	0.0384±0.0035	0.0074±0.0027

ResNet-50 with 92.15%. However, within the CNN-SVM hybrid models, the highest accuracy was obtained from the AlexNet hybrid without LASSO, reaching 97.37% on this dataset.

Moreover, across all hybrid configurations—regardless of whether LASSO was used—the execution time was significantly shorter than that of the conventional CNN-based classification processes.

Beyond overall accuracy, the F1-score results across all datasets reveal an important advantage of the proposed hybrid models. In several cases, conventional CNNs achieved relatively high accuracy values but exhibited considerably lower F1-scores, indicating poor performance on minority classes. In contrast, the hybrid CNN-SVM models, consistently improved the F1-scores alongside accuracy. This improvement can be attributed to the margin-based optimization of the SVM classifier which contrasts with the probabilistic output of the softmax layer in conventional CNNs. By constructing decision boundaries that maximize

TABLE 14. Fabric dataset results for MobileNet V2.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.9160	0.9677±0.0009	0.9685±0.0011
F1	0.3260	0.4285±0.0127	0.4113±0.0125
PRE	0.3574	0.3884±0.0123	0.3703±0.0108
SENS	0.4490	0.5263±0.0179	0.5582±0.0393
SPE	0.9765	0.9557±0.0043	0.9747±0.003
Time	3.0866	0.0253±0.0029	0.0065±0.0004

TABLE 15. Fabric dataset results for ShuffleNet.

	CNN	CNN-SVM Hybrid Without LASSO	CNN-SVM Hybrid With LASSO
ACC	0.9144	0.9696±0.0012	0.9692±0.001
F1	0.3356	0.4814±0.0235	0.4415±0.0137
PRE	0.3581	0.4288±0.0186	0.3825±0.011
SENS	0.4476	0.6121±0.0354	0.6602±0.0298
SPE	0.9652	0.9617±0.0035	0.9788±0.0024
Time	2.4413	0.0125±0.0027	0.0048±0.0003

class separation, SVMs better capture minority classes, leading to more balanced classification outcomes. Consequently, the hybrid CNN-SVM framework not only enhances classification accuracy but also achieves significantly higher F1-scores.

In general, the AlexNet-SVM without LASSO method yielded the best classification performance across all datasets. For each dataset, the confusion matrix constructed from the metric values closest to the average obtained over 30 runs with this method is presented as an example in Figures 7, 8, and 9.

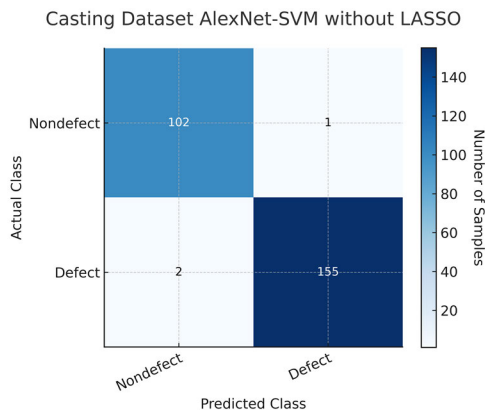


FIGURE 7. Casting dataset AlexNet-SVM without LASSO confusion matrix.

D. PERFORMANCE COMPARISON AND STATISTICAL ANALYSIS

In this study, the classifier layers of four CNN architectures were replaced with an SVM classifier. To reduce the

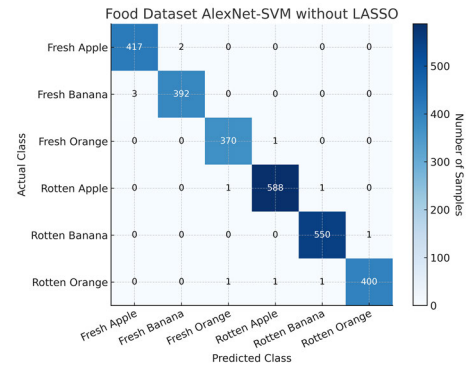


FIGURE 8. Food dataset AlexNet-SVM without LASSO confusion matrix.

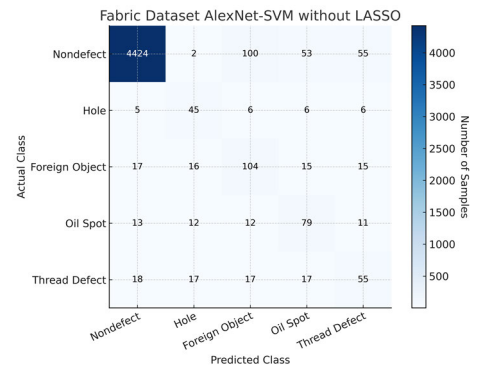


FIGURE 9. Fabric dataset AlexNet-SVM without LASSO confusion matrix.

high-dimensional feature space generated by CNN-based feature extraction, the LASSO feature selection method was applied. The use of an SVM classifier layer led to improved classification performance compared to conventional CNN architectures. Moreover, incorporating LASSO significantly reduced model complexity and improved execution time, enhancing overall efficiency. These enhancements represent important steps toward improving the speed performance metric, which is critical for real-time applications.

1) PERFORMANCE COMPARISONS WITH LITERATURE STUDIES

Numerous studies have been conducted to detect manufacturing defects using various datasets and methods reported in the literature. A comparison of classification performance with previous state-of-the-art approaches on these datasets is presented in Tables 16-18. The method achieving the highest accuracy is highlighted in bold. Accuracy comparisons were made with studies in the literature. However, it is important to note that direct performance comparisons may be affected by subtle differences in experimental conditions (e.g., data pre-processing, specific training/testing splits, or hyperparameter tuning) that are not always reported.

When previous studies using the Casting dataset were examined, the highest classification accuracy was achieved with Smart Quality Inspection method [62]. Among the

TABLE 16. Casting dataset literature Comparison.

Author	Method	Accuracy
Ekambaram et al. [59]	DenseNet	99.42%
Benbarrad et al. [60]	EfficientNetB0	96.88%
Tsiktsiris et al. [61]	Accelerated CNN	99.72%
Sundaram et al. [62]	Smart Quality Inspection	99.86%
Lilhore et al. [63]	VGG-16	95.80%
Nguyen et al. [64]	Transfer Learning DenseNet	95.94%
Proposed	AlexNet-SVM without LASSO	98.29%±0.0091

TABLE 17. Food dataset literature comparison.

Author	Method	Accuracy
Karakaya et al. [65]	ResNet50 Features	96.72%
Mishra et al. [66]	VGG-16	97.5%
Miah et al. [67]	Inception V3	97.34%
Chakraborty et al. [68]	MobileNetV2	99.61%
Kazi et al. [69]	ResNet 50	99.70%
Proposed	AlexNet -SVM without LASSO	99.82%±0.0003

TABLE 18. Fabric dataset literature comparison.

Author	Method	Accuracy
Arshad et al. [70]	VGG16	73.91%
Ciklacandir et al. [71]	ResNet18 Ensemble Learning	95.32%
Guder et al. [72]	Majority Voting	93.41%
Wan et al. [73]	High-Frequency Features	78.80%
Kure et al. [74]	Local Homogeneity Analysis	96.40%
Proposed	AlexNet-SVM-without LASSO	97.37% ± 0.001

hybridization methods proposed in this study, the highest accuracy was obtained using the AlexNet-SVM hybrid without LASSO, yielding $98.29\% \pm 0.0091$. This result is above the average of existing methods in the literature and is very close to the best classification performance.

For the Food and Fabric datasets, the highest classification accuracy among the four CNN-SVM hybrid models proposed was also achieved by the AlexNet-SVM model without LASSO, similar to the results obtained for the Casting dataset, with accuracies of $99.82\% \pm 0.0003$ and $97.37\% \pm 0.001$, respectively. This proposed method is expected to contribute significantly to the literature for both datasets.

The CNN-SVM hybridization approach, designed for the detection of manufacturing defects, demonstrated very high classification performance across all datasets. While the hybrid approach involving AlexNet consistently outperformed others, the use of LASSO also led to notable improvements in classification speed.

This study explores the hybridization of Support Vector Machine (SVM) classifiers with four Convolutional Neural Network (CNN) architectures, incorporating the Least Absolute Shrinkage and Selection Operator (LASSO) for feature selection to address the challenge of high-dimensional feature spaces. The proposed hybrid approach demonstrated improved classification performance compared to conventional CNN architectures. Notably, the AlexNet-SVM hybrid

TABLE 19. Test of normality.

Dataset Name	CNN Architecture	Kolmogorov-Smirnov Significance		Shapiro-Wilk Significance	
		Without LASSO	With LASSO	Without LASSO	With LASSO
Casting	AlexNet	0.200	0.065	0.550	0.056
	ResNet 50	0.121	0.06	0.197	0.169
	MobileNet V2	0.200	0.200	0.572	0.348
	ShuffleNet	0.200	0.196	0.846	0.180
Food	AlexNet	0.103	0.200	0.637	0.990
	ResNet 50	0.200	0.200	0.597	0.983
	MobileNet V2	0.200	0.200	0.324	0.233
	ShuffleNet	0.200	0.200	0.286	0.763
Fabric	AlexNet	0.119	0.200	0.053	0.582
	ResNet 50	0.200	0.200	0.789	0.685
	MobileNet V2	0.200	0.062	0.056	0.156
	ShuffleNet	0.174	0.200	0.101	0.708

TABLE 20. All conventional CNN vs CNN-SVM hybrid one sample t-test results.

		Casting	Food	Fabric
AlexNet vs AlexNet-SVM	Without LASSO	2.9991E-35	1.5872E-159	6.6917E-146
	With LASSO	1.6219E-31	3.5514E-157	3.0033E-146
ResNet vs ResNet 50-SVM	Without LASSO	2.7287E-35	8.59E-158	2.2175E-145
	With LASSO	1.8105E-34	5.388E-153	8.2953E-146
MobileNet V2 vs MobileNet V2-SVM	Without LASSO	0.0059	1.9842E-144	1.6171E-146
	With LASSO	0.000001	1.1884E-142	2.3299E-144
ShuffleNet vs ShuffleNet-SVM	Without LASSO	4.1022E-16	2.5758E-147	6.0743E-143
	With LASSO	3.485E-20	1.4735E-143	3.9988E-146

without LASSO achieved the highest classification accuracies: $98.29\% \pm 0.0091$ for the Casting dataset, $99.82\% \pm 0.0003$ for the Food dataset and $97.37\% \pm 0.001$ for the Fabric dataset. These results highlight significant improvements in both classification accuracy and computational efficiency, particularly in the context of manufacturing defect detection. This research contributes to the existing literature by demonstrating the effectiveness of CNN-SVM hybridization across diverse datasets and provides valuable insights for future studies in this domain.

2) STATISTICAL ANALYSIS

Within the scope of this study, a one sample t-test was conducted to determine whether the differences between the CNN-SVM hybrid and conventional CNN models, as well as between CNN-SVM hybrid and prior studies in the literature, were statistically significant. To perform this analysis, 30 classification accuracy values from repeated executions were tested for normality using IBM SPSS. The Shapiro

TABLE 21. Casting dataset one-sample t-test p values.

	AlexNet-SVM		ResNet 50-SVM		MobileNet V2-SVM		ShuffleNet-SVM	
	Without LASSO	With LASSO	Without LASSO	With LASSO	Without LASSO	With LASSO	Without LASSO	With LASSO
DenseNet [59]	1.8617E-7	3.2323E-13	2.1593E-19	1.2388E-22	1.11E-36	3.00E-34	3.1485E-30	5.7674E-32
EfficientNetB0 [60]	2.0706E-9	4.08E-12	3.6227E-12	3.6802E-17	1.04E-35	3.41E-33	1.0891E-28	2.4833E-30
Accelerated CNN [61]	1.7835E-9	1.8924E-14	4.8418E-20	3.7504E-23	8.65E-37	2.28E-34	2.1283E-30	3.8121E-32
Smart Quality Inspection [62]	2.3535E-10	5.4355E-15	2.4689E-20	2.1818E-23	7.69E-37	2.01E-34	1.776E-30	3.1485E-32
VGG-16 [63]	3.0177E-15	0.000004	1.1179E-7	5.9702E-14	2.86E-35	1.02E-32	5.6519E-28	1.4418E-29
Transfer Learning DenseNet [64]	1.3274E-14	0.000031	2.5555E-8	2.0758E-14	2.50E-35	8.85E-33	4.5415E-28	1.141E-29

TABLE 22. Food dataset one-sample t-test p values.

	AlexNet-SVM		ResNet 50-SVM		MobileNet V2-SVM		ShuffleNet-SVM	
	Without LASSO	With LASSO	Without LASSO	With LASSO	Without LASSO	With LASSO	Without LASSO	With LASSO
ResNet50 Features [65]	1.05E-58	4.33E-56	1.39E-56	1.82E-51	1.90E-35	2.85E-33	1.74E-40	1.03E-35
VGG-16 [66]	4.76E-55	2.41E-52	9.26E-53	1.58E-47	7.48E-27	2.66E-24	8.36E-34	2.71E-28
Inception V3 [67]	6.86E-56	3.29E-53	1.21E-53	1.92E-48	3.77E-29	1.01E-26	1.71E-35	3.30E-30
MobileNetV2 [68]	9.47E-25	5.23E-18	2.94E-13	2.15E-01	2.48E-33	4.94E-32	1.66E-32	1.66E-30
ResNet 50 [69]	7.42E-18	9.96E-08	7.77E-01	2.80E-08	3.71E-34	7.95E-33	1.39E-33	1.90E-31

TABLE 23. Fabric dataset one-sample t-test p values.

	AlexNet-SVM		ResNet 50-SVM		MobileNet V2-SVM		ShuffleNet-SVM	
	Without LASSO	With LASSO	Without LASSO	With LASSO	Without LASSO	With LASSO	Without LASSO	With LASSO
CNN VGG16 [70]	6.38E-71	3.47E-71	1.09E-69	1.28E-70	3.63E-71	4.71E-69	1.01E-67	7.01E-71
ResNet18 Ensemble Learning [71]	3.21E-40	1.39E-39	3.21E-30	5.99E-39	2.04E-36	5.88E-35	1.70E-34	2.57E-37
Majority Voting [72]	1.64E-48	2.33E-48	7.17E-44	9.23E-48	5.25E-47	3.75E-45	3.51E-44	3.36E-47
High-Frequency Features [73]	5.61E-68	3.21E-68	1.39E-66	1.19E-67	3.90E-68	4.93E-66	1.01E-64	7.16E-68
Local Homogeneity Analysis [74]	8.02E-31	5.32E-29	2.00E-06	2.82E-28	1.96E-19	8.55E-20	2.94E-21	2.58E-23

Wilk and Kolmogorov Smirnov tests, both nonparametric tests for assessing normal distribution, were employed for this purpose. If the significance (p-value) obtained from either test exceeded the predetermined threshold (alpha), the null hypothesis (H_0) which assumes that the data follow a normal distribution was accepted. Conversely, if the p-value was less than the alpha level, the null hypothesis was rejected, indicating that the data did not follow a normal distribution.

The hypotheses for the normality test are defined as follows:

H_0 (Null Hypothesis): The classification accuracy results obtained from 30 executions of the dataset follow a normal distribution.

H_1 (Alternative Hypothesis): The classification accuracy results obtained from 30 executions of the dataset do not follow a normal distribution.

In this study, the significance level (alpha) was set at 0.05. The p-values obtained for each dataset are presented in Table 19.

Considering the p-values obtained from the 30 execution results across the three datasets in all hybrid studies, it was observed that all values were greater than 0.05. As a result,

the data were determined to follow a normal distribution. In this case, the null hypothesis (H_0) was accepted, and the alternative hypothesis (H_1) was rejected.

Following the confirmation of normality, a one-sample t-test was conducted to determine whether there was a statistically significant difference between the classification accuracies obtained from 30 executions of the proposed CNN-SVM hybrid model and those of the conventional CNN models for each dataset. The results of the one-sample t-tests comparing conventional CNNs with the CNN-SVM hybrids are presented in Table 20. This table illustrates whether the differences in test accuracies between the proposed hybrid CNN-SVM and the conventional CNN models are statistically significant.

The hypotheses for the t-test at a significance level of $\alpha = 0.05$ are defined as follows:

H_0 (Null Hypothesis): There is no statistically significant difference between the test accuracies of the proposed hybrid CNN-SVM model and the conventional CNN algorithms.

H_1 (Alternative Hypothesis): There is a statistically significant difference between the test accuracies of the proposed hybrid CNN-SVM model and the conventional CNN algorithms.

According to t-test results obtained from 30 executions results of datasets, the p values were found to be less than 0.05 at a 95% confidence level. Consequently, the null hypothesis (H_0), which states that there is no statistically significant difference between the proposed CNN-SVM hybrid and the conventional CNN models, was rejected. Instead, the alternative hypothesis (H_1), which asserts that there is a statistically significant difference between the proposed CNN-SVM hybrid and conventional CNN models, was accepted.

Furthermore, a similar analysis was conducted to determine whether there is a significant difference between the classification accuracies obtained from the 30 executions of the proposed hybrid CNN-SVM algorithm and those reported in other CNN studies in the literature. The hypotheses for this t-test at the $\alpha = 0.05$ significance level are as follows:

H_0 (Null Hypothesis): There is no statistically significant difference between the proposed CNN-SVM hybrid and other methods reported in the literature.

H_1 (Alternative Hypothesis): There is a statistically significant difference between the proposed CNN-SVM hybrid and other methods reported in the literature.

Considering the test accuracy values for each datasets; for the Casting dataset, a significant difference was observed for all proposed CNN-SVM hybrids, both with and without LASSO. The corresponding p-values are presented in Table 21, with statistically significant differences favouring the proposed hybrid method highlighted in bold. For the Food dataset, no significant difference was found between ResNet-50 and the proposed ResNet-50-SVM hybrid without LASSO, nor between MobileNetV2 and the proposed ResNet-50-SVM hybrid with LASSO. However, significant differences were observed for all other proposed CNN-SVM hybrids, both with and without LASSO. The p-values for these comparisons are shown in Table 22, with significant differences in favour of the proposed hybrid method indicated in bold. For the Fabric dataset, significant differences were found for all proposed CNN-SVM hybrids, both with and without LASSO. The relevant p-values are provided in Table 23, with significant differences supporting the proposed hybrid method highlighted in bold.

This study employs a one-sample t-test to evaluate the statistical significance of differences between the proposed Convolutional Neural Network-Support Vector Machine (CNN-SVM) hybrid and both conventional CNN architectures and prior studies reported in the literature. Thirty classification accuracy results were first tested for normality using IBM SPSS, with the Shapiro–Wilk and Kolmogorov–Smirnov tests confirming that the data conform to a normal distribution ($p > 0.05$). Subsequently, the t-test results revealed statistically significant differences ($p < 0.05$) in test accuracies between the CNN-SVM hybrid and conventional CNN models, leading to the rejection of the null hypothesis. Additionally, comparisons with other CNN methods demonstrated significant differences for the Casting and Fabric datasets,

whereas some comparisons within the Food dataset did not yield statistically significant differences.

V. CONCLUSION

In this study, a hybrid model combining Convolutional Neural Networks (CNN) and Support Vector Machines (SVM) is proposed to enhance defect detection across various product types. The presented approach effectively integrates the feature extraction capabilities of CNNs with the classification strengths of SVMs, addressing the challenges associated with high-dimensional feature spaces. The backbone CNN architectures were employed to extract deep features from input images, while the Least Absolute Shrinkage and Selection Operator (LASSO) feature selection method was used to identify the most relevant features. This process reduced the dimensionality of the high-dimensional feature maps obtained from CNNs, thereby lowering computational complexity. As a result of the reduced execution time achieved through the use of LASSO feature selection, a fast and efficient hybrid algorithm has been developed with the capability to operate in real-time applications. This algorithm can be employed for the detection and classification of surface defects in products with different surface types in manufacturing systems.

Experimental results demonstrated that the proposed hybrid model significantly outperformed traditional CNN architectures in both classification accuracy and execution time. Specifically, the AlexNet-SVM hybrid model without the LASSO feature selection method achieved the highest accuracy rates of 98.29%, 99.82%, and 97.37% for the casting, food, and fabric datasets, respectively. These results indicate a substantial improvement over conventional methods, highlighting the potential of hybrid models for real-time defect detection. Moreover, incorporating the LASSO feature selection method proved effective in reducing model complexity and enhancing computational efficiency. Statistical analyses further confirmed the robustness of the findings, revealing significant differences in performance metrics between the CNN-SVM hybrid and conventional CNN models.

In addition to improvements in accuracy and execution time, the proposed hybrid CNN-SVM models also achieved higher F1-scores across all datasets. This indicates that the hybrid approach provides more balanced classification performance, particularly by more effectively capturing minority classes. Such robustness is crucial in industrial defect detection, where failing to identify rare but critical defect types can lead to significant quality degradation and increased costs.

For future research, hybridizing custom CNN architectures for feature extraction may enable even faster and more accurate detection of various manufacturing defects. Such advancements could contribute to increased production efficiency and reduced operational costs in industrial manufacturing processes.

VI. DECLARATIONS

A. CONFLICT OF INTEREST

The authors declare that there are no potential conflicts of interest regarding any financial support, research, authorship, and publication of this article.

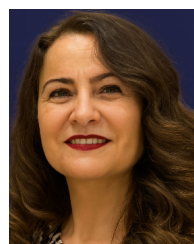
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